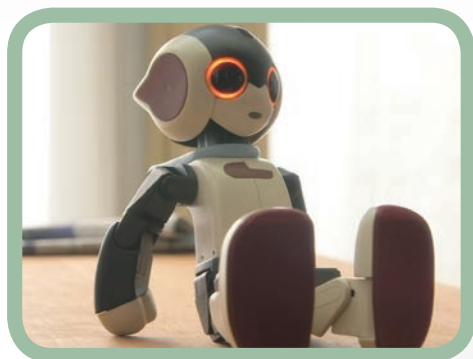
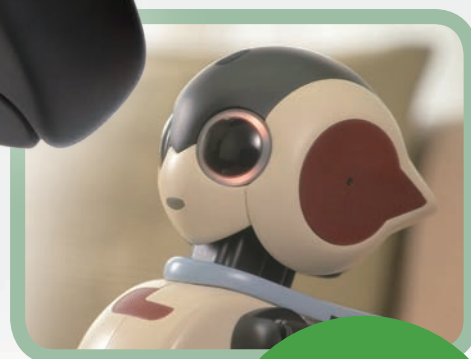


Build your own **Robi**

Pack 08



Stages 27-30:
Begin building up
Robi's right leg

Build your own **Robi** CONTENTS

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Stage 30: Assemble Robi's right knee	p150

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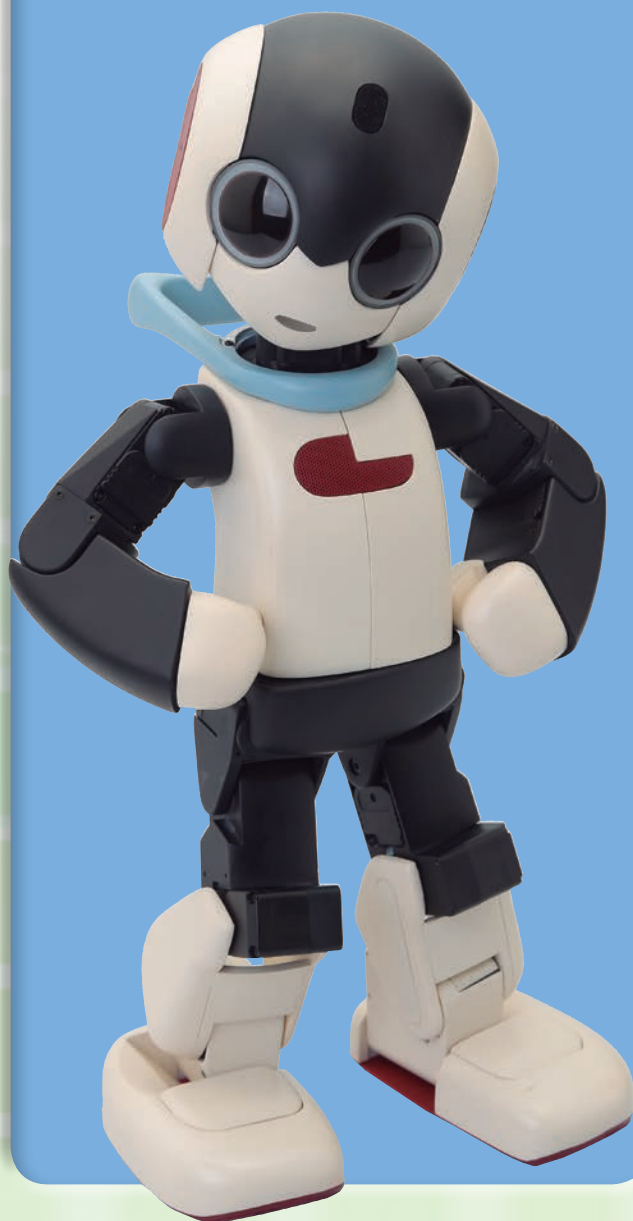
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STAGE 27: CONTINUE BUILDING UP ROBI'S RIGHT FOOT

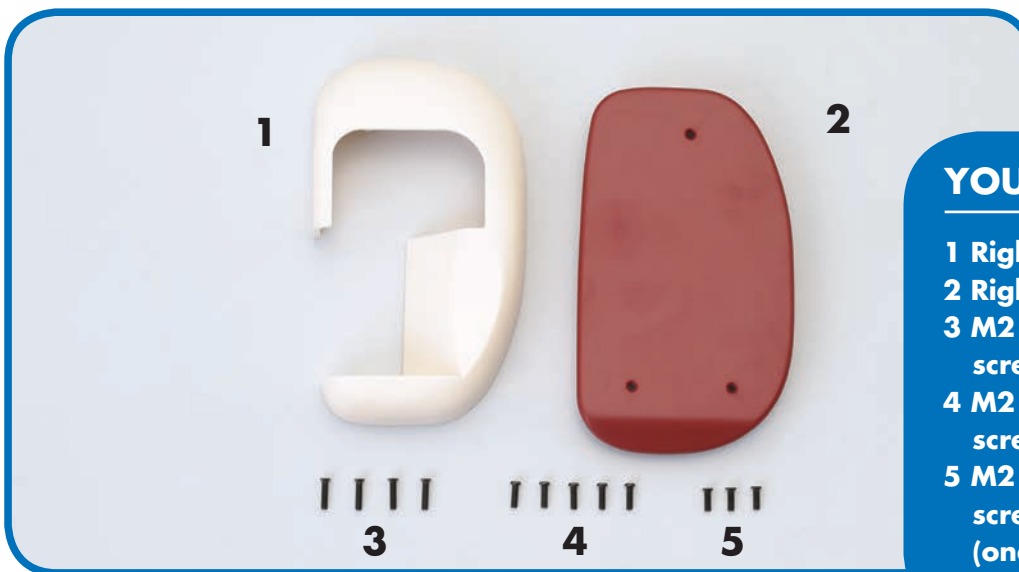
In this session, you will build up Robi's right foot by adding the ankle servo, prepared over the previous stages, to the rest of his right foot.

This session sees you joining up two of Robi's servos again, in this case his right foot and ankle servos. You will fit both servos to their respective frames in one assembly session, so be sure to follow the steps closely to avoid any

confusion. You will also be handing servo cables and opening servos, so, as always, take great care not to touch, disturb or damage any of the delicate electronic components during the assembly process.



PARTS TO BE ASSEMBLED



YOUR PARTS

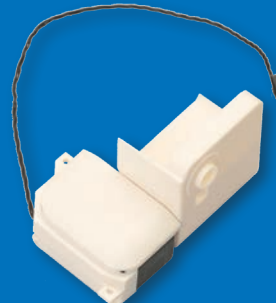
- 1 Right foot side cover
- 2 Right sole
- 3 M2 x 7mm countersunk screws x 4
- 4 M2 x 6mm button-head screws x 5
- 5 M2 x 6mm countersunk screws x 3 (one of each screw is a spare)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Taped and wired ankle servo (Stage 26)

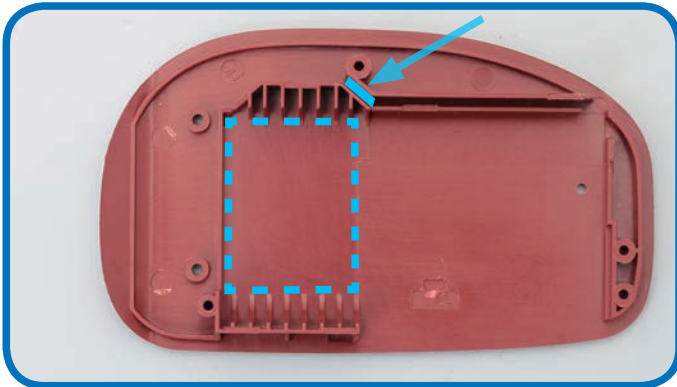


Foot servo and ankle frame (Stage 25)

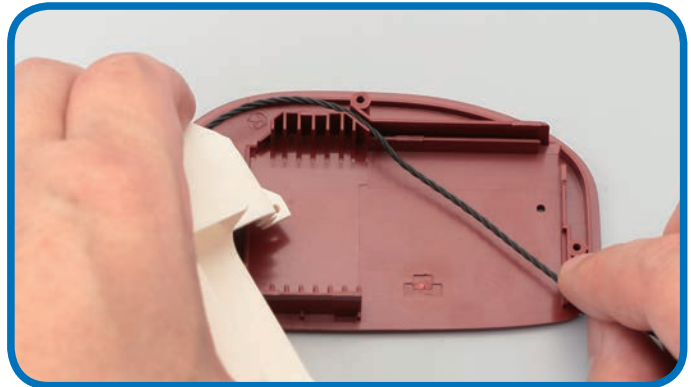


Right ankle side panel and joint (Stage 24)

FITTING THE FOOT SERVO



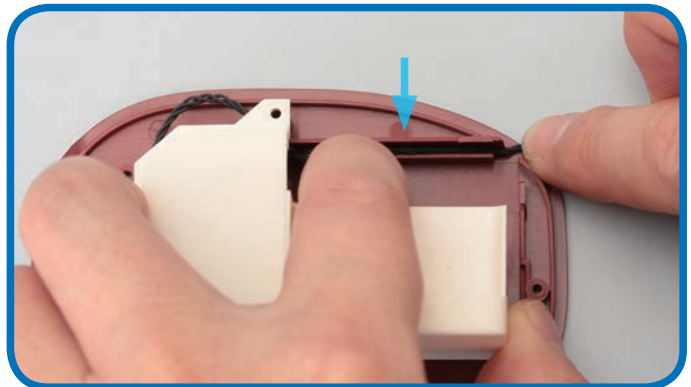
- 1** Familiarise yourself with the inside of the right sole. The first servo you will fit in this stage will be placed in the area marked with blue dashed lines. The angled groove (arrowed) will accommodate the servo's cable once fitted.



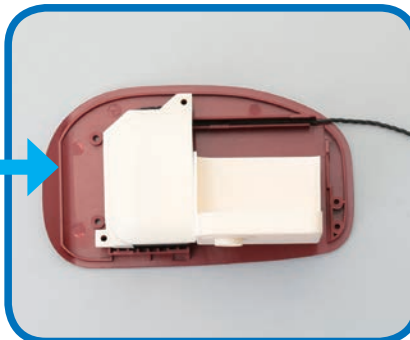
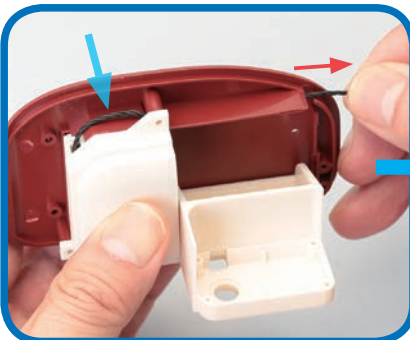
- 2** Take the foot servo and ankle frame (Stage 25) in one hand, and with the other feed the servo cable through the angled groove highlighted in the previous step.



- 3** Making sure the cable does not come out of the groove, lower the servo, positioned as shown, into the rectangular space highlighted in Step 1. Pull the cable gently to the right as you do this to minimise slack.



- 4** Fit the length of cable shown into the groove.



- 5** Now you need to eliminate the slack at the servo end completely, which is done by gently pulling the free end of the cable through the groove (hold the cable itself – do not pull on the connector). Be very careful not to tug on the cable too hard, as this could damage it, or cause it to come free inside the servo, which will mean opening the servo and re-securing the cable. If the cable comes out of the groove as you pull it, simply push it back in.



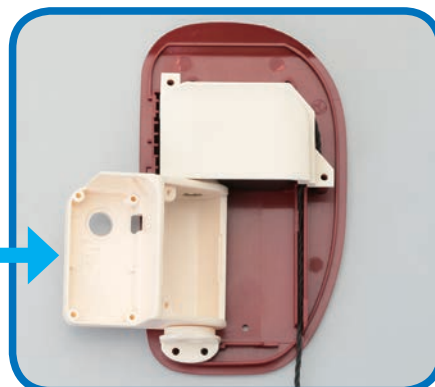
- 6** Hold the foot servo and its cover in place, then tilt the ankle frame over to expose its inside, as shown.



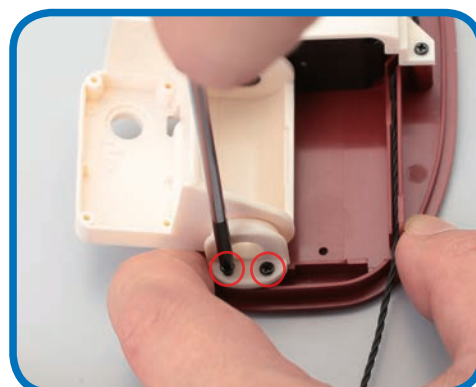
- 7** Take the right ankle joint saved from Stage 24, and line it up to the circled hole at the rear face of the opened ankle frame. Orientate the part as shown.



- 8** Hold the ankle frame in place with one hand, and with the other push the joint into the hole so that your assembly looks like the one in the right-hand photo above.



- 9** Still holding the ankle frame in position, insert and tighten two M2 x 6mm button-head screws into the circled holes to secure the foot cover and ankle frame sub-assembly to the right sole.



- 10** Now tighten two more M2.6 x 6mm button-headed screws into the circled holes on the ankle joint, securing this to the sole.

FITTING THE ANKLE SERVO



- 11** Take the free end of the servo cable fitted into the sole's side groove in Step 4, and feed it around and into the hole in the ankle joint.



- 12** Now pull the cable through the inside of the ankle frame and pass it out through the square hole in the frame's side.

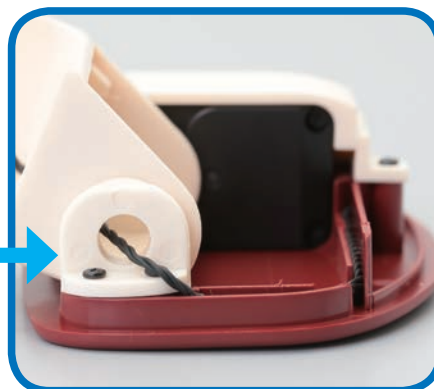




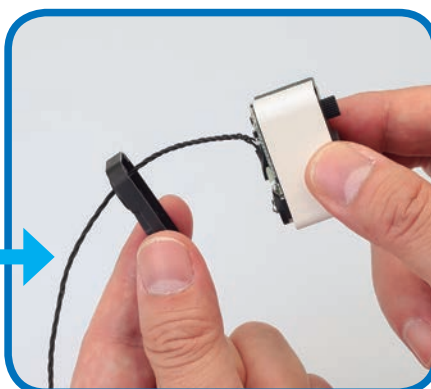
13 Now fit the indicated section of the cable into the groove at the heel of the sole.



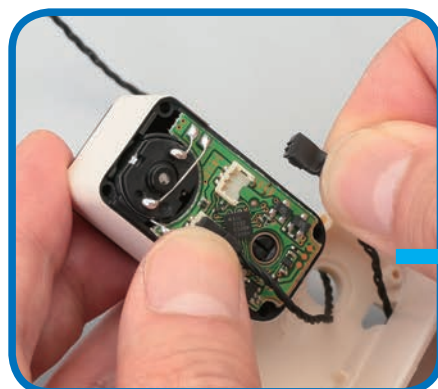
14 Your assembly should look like this, with the cable sitting neatly inside the grooves at the side and heel of the sole, and running with a minimum of slack into the ankle frame.



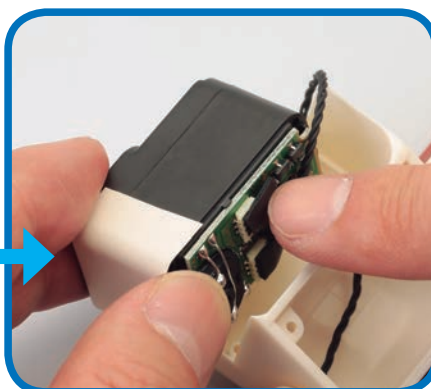
15 Next, take the wired and taped ankle servo (Stage 26) and remove the four long screws holding its back cover in place. Keep the screws safely until you need them again, in Step 21. Remove the cover, without disturbing the cable fitted to the servo's circuit board.



16 Lay the servo face up, next to the ankle frame and sole assembly. Feed the free end of the ankle servo's cable over and through the circular hole in the ankle frame **from the inside of the part**.



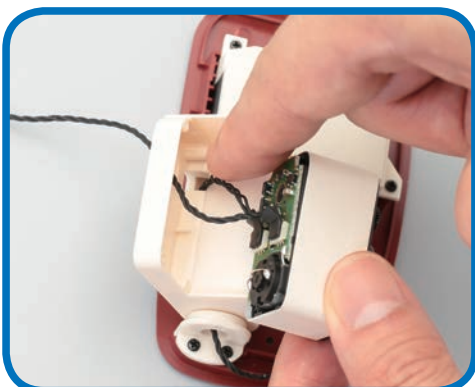
17 Hold the open ankle servo in your hand and very carefully attach the connector of the foot servo's cable (running from the square hole in the ankle frame) to its free connector.



TIP!

If you did not fit the foot servo's cable with its protective pad when wiring the servo previously, do so now. As ever, be very careful not to touch any of the delicate exposed electronic parts as you do this.

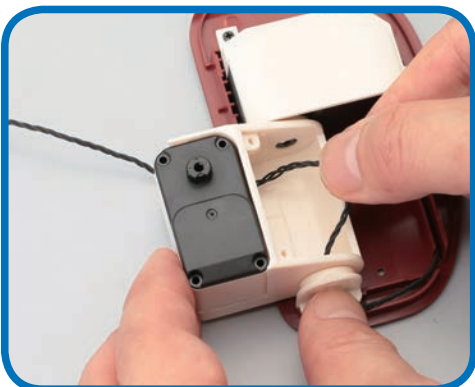
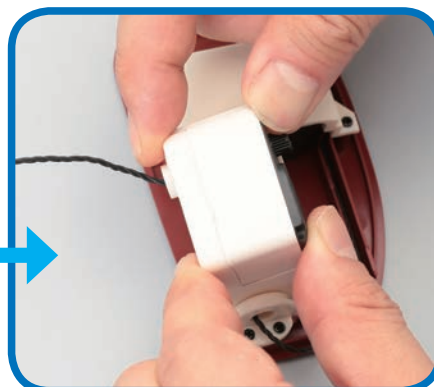




- 18** Lower the ankle servo, orientated as shown, into the ankle frame, pushing the servo cable running from the square hole back through the hole so that there is no wire bunching up.



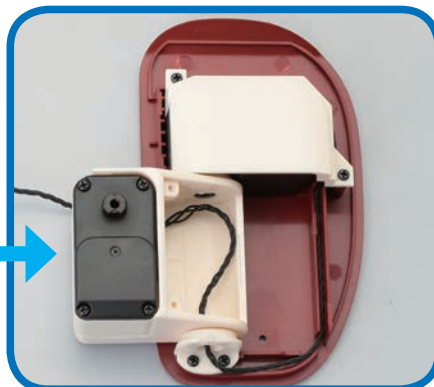
- 19** Pulling the ankle servo's cable through the frame's circular hole, again to prevent the cable bunching up inside the frame, press the servo into the frame so that it fits snugly. There should be no gaps visible between the edge of the ankle frame and the white tape wrapped around the servo.



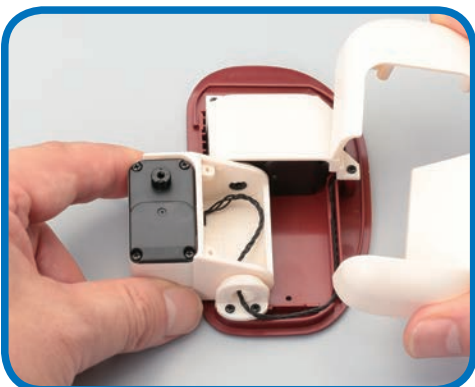
- 20** Gently pull the foot servo's cable further through the frame's square hole, ensuring there is no excess left between the ankle servo's circuit board and the frame's inside edge.



- 21** When you are happy that there is no cable bunched up inside the frame (both cables should be running neatly out of their respective holes), use the four screws removed in Step 15 to secure the servo to the ankle frame.



FITTING THE FOOT SIDE COVER



- 22** Line up the foot side cover to your assembly – you will see how the part is cut to fit neatly on top of the various parts of Robi's foot and ankle.



- 23** Lower the foot side cover onto the assembly. Be careful not to catch any of the cables as you do so, especially those around the heel.

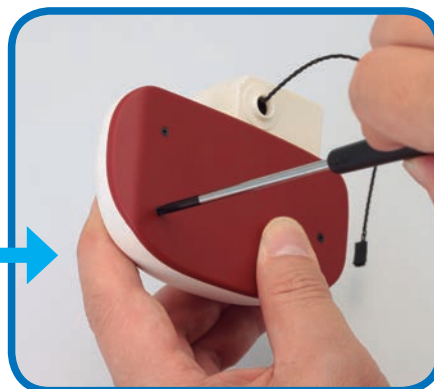




- 24** Making sure that none of the cable is caught, press the foot side cover into place.



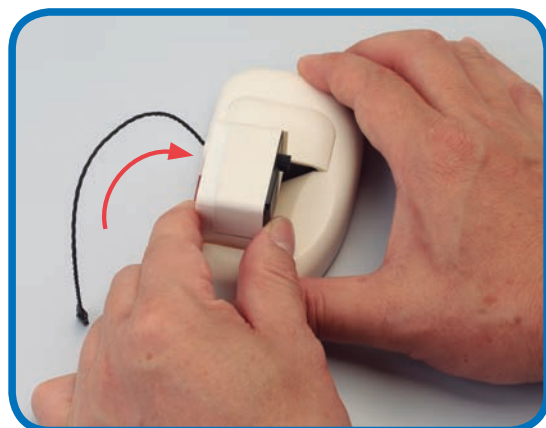
- 25** Holding the parts together firmly so they do not get disturbed, turn the whole assembly over so you can access the sole. Insert and tighten three M2 x 7mm countersunk screws into the circled holes to secure the sole to the rest of the foot.



- 26** Lay Robi's right foot assembly the right way up on your work surface, then fit the ankle side panel (Stage 24) to the open section of the ankle frame.



- 27** Use two M2 x 6mm countersunk screws to secure the part.



- 28** Rotate Robi's ankle frame upwards to complete this assembly session.

NOTE: As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

Assembled right foot

At the completion of this stage, Robi's right foot has taken shape, with all the foot covers in place and the foot and ankle servos linked, ready for the next phase of the build.



STAGE 28: BEGIN BUILDING UP ROBI'S RIGHT LOWER LEG

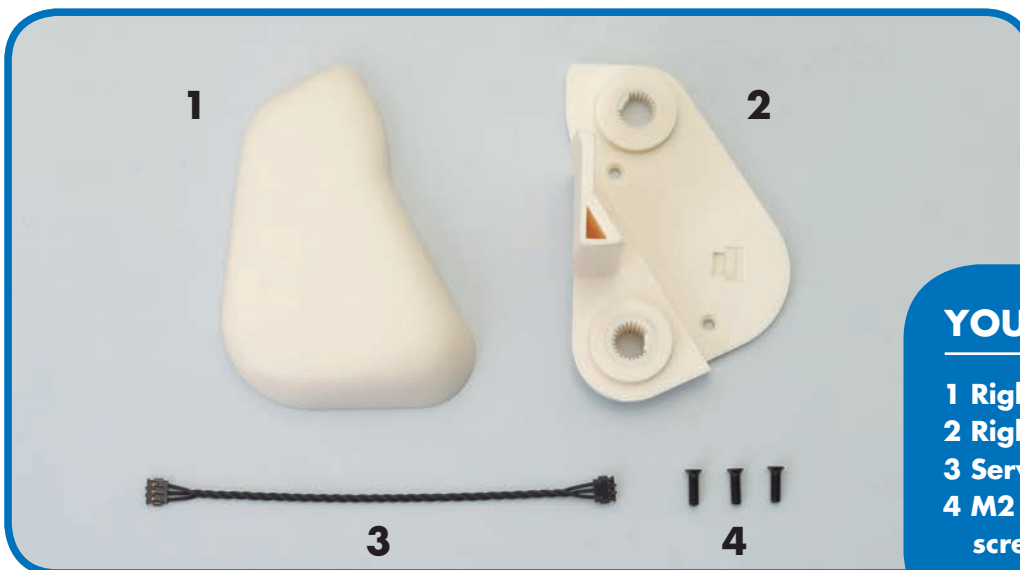
In this stage, you will begin to assemble Robi's right lower leg. You start with the lower leg frame, which will join the ankle and knee servo, and then add the outer cover.

Now that you have fitted the foot and ankle servos into their respective frames, completing the assembly of Robi's right foot, it is time to add the parts that will connect his foot to the rest of his leg. To do this, you will fit

the lower leg frame that will join Robi's ankle servo to his knee servo, before adding the outer lower leg cover. Remember to take care when fitting the servo shaft into the D-cut hole in the lower leg frame.



PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Right outer lower leg cover
- 2 Right lower leg frame
- 3 Servo cable (70mm)
- 4 M2 x 6mm countersunk screws x 3

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.

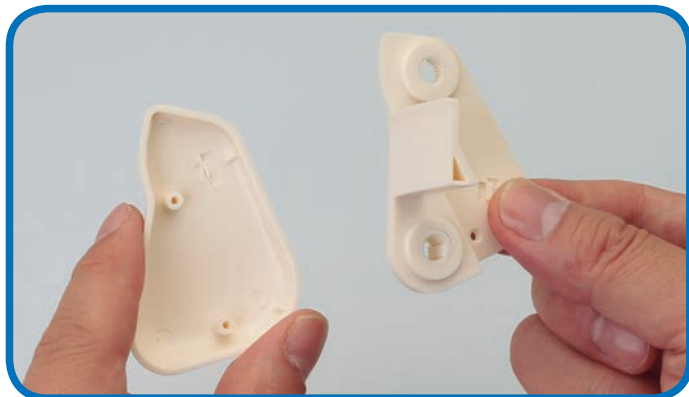


*Right foot assembly
(Stage 27)*



*Protective pads
(Stage 3)*

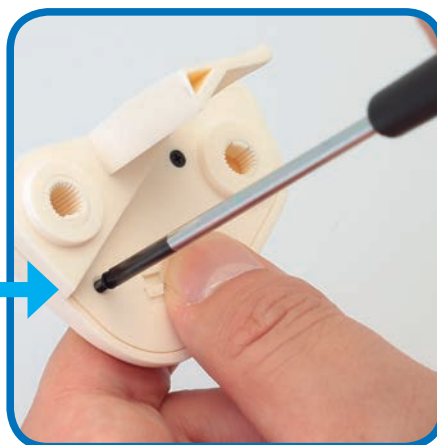
FITTING THE LOWER LEG FRAME



1 Line up the lower leg frame with its outer cover, following the orientation shown in the photo exactly.



2 Press the parts together, as shown. They should fit neatly into one another.



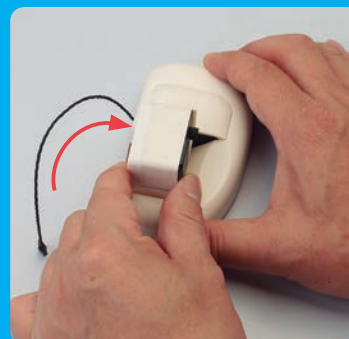
3 Use two M2 x 6mm countersunk screws to secure the parts.

Note the D-cut holes seen on the inside of the lower leg frame – the lower hole will be the one into which you will fit the corresponding servo shaft in this stage.



TIP!

Robi's ankle will need to be fully upright and perpendicular to his foot for you to fit the lower leg frame easily. Make sure it is in this position before you continue with this stage's build.

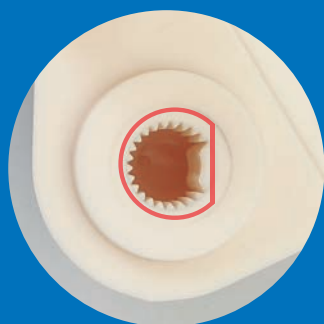


4 Take the right foot assembly (Stage 27) and line up the lower leg frame with its ankle servo and frame. The front panel, Robi's 'shin', will stretch across the forward-facing side of the servo.

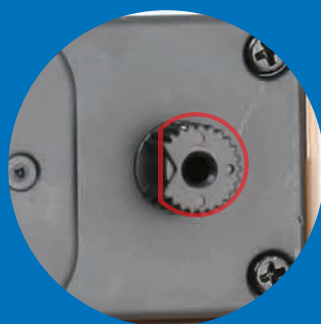
THE D-SHAPED HOLE



It is very important that you take care when working with the servos. Each joint, such as the lower leg frame you will be fitting in this stage, has a precisely moulded D-shaped hole, designed to fit snugly onto the D-shaped servo shaft to ensure that the servo is set correctly. When these parts are aligned as they should be, it will be easy to fit them together. If it is in any way difficult to join the parts, separate them, check that you have both of them lined up accurately, and try again. Never force the parts together, as this could damage both of them.



The D-shaped hole in the lower leg frame, supplied with this stage.

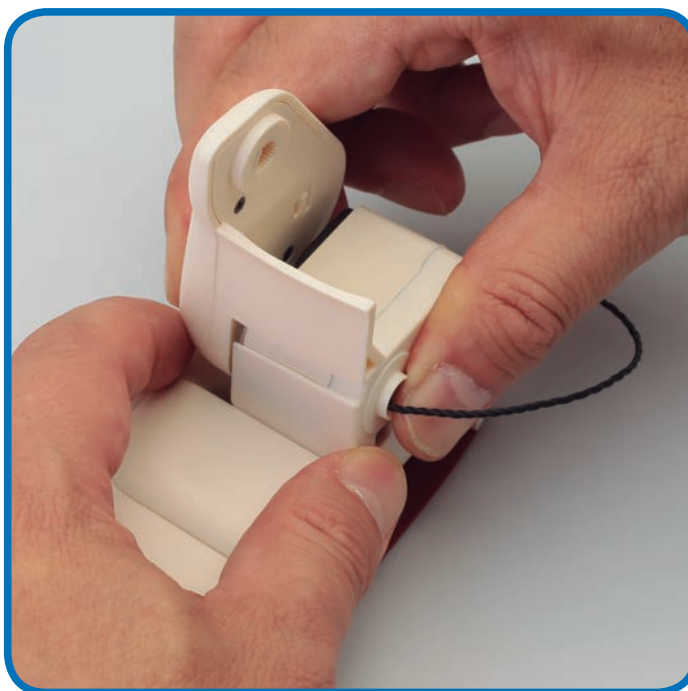


The servo's shaft is also D-shaped, and is designed to fit snugly into the hole in the frame.

Never force the parts together, because there is a high risk that you will damage the small plastic teeth inside the hole, as seen below.

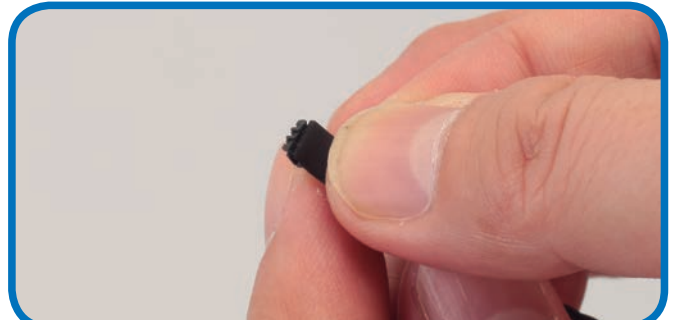
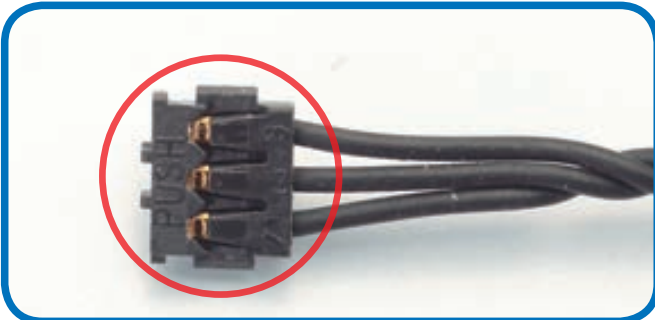
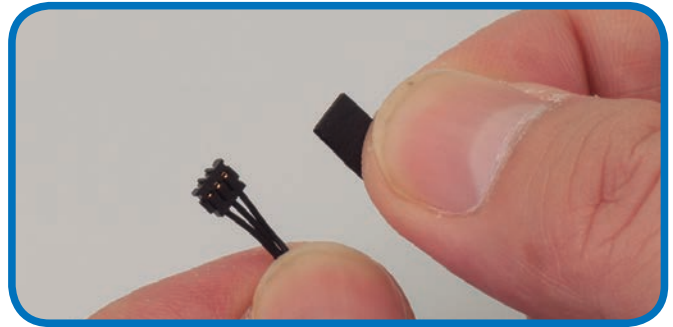
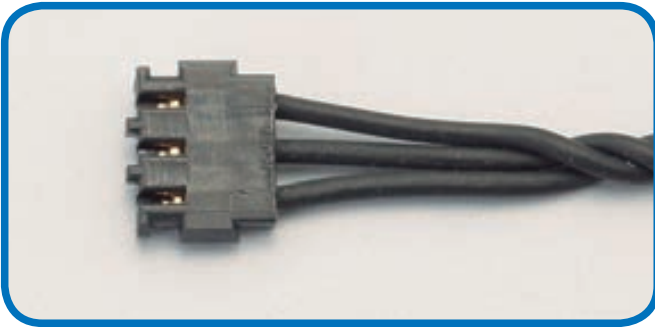


- 5** Slide the lower leg frame over the front of the right ankle servo and frame. The leg frame's lower hole should fit onto the servo shaft. **Do not force the parts together – see the box above.**



- 6** Once you are certain that the D-shaped hole and servo shafts are aligned correctly, grip the parts between your fingers and thumbs, and press them together. Again, be very careful not to force them.

SEALING THE SERVO CABLE



7 As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).

8 Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

It is advisable to fit the servo cable's protective pads at this stage, but if you want to wait until the cable is used in the build, you can do so. In that case, simply store the cable away safely until it is needed, and remember to apply the pads then.



Assembled right foot and lower leg

At the completion of this stage, Robi's right foot has taken shape, with all the foot covers in place and the foot and ankle servos linked, ready for the next phase of the build.



STAGE 29: PREPARE ROBI'S RIGHT KNEE SERVO

In this assembly, you will prepare, test and set the servo motor that will operate Robi's right knee – servo ID=4.

Now that you have wired, programmed and fitted the servos that will operate Robi's right foot and ankle (remember, now that the servos are linked by their cables, you must **not** attach them to the test board!), it is time to start

working further up his leg to fit his right knee servo. The work is the same as for his other servos, but it is vital to take great care with it because it involves exposing the delicate electronic parts.



YOUR PARTS

1 Servo motor

PARTS TO BE
ASSEMBLED



SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



70mm servo cable
(Stage 28)



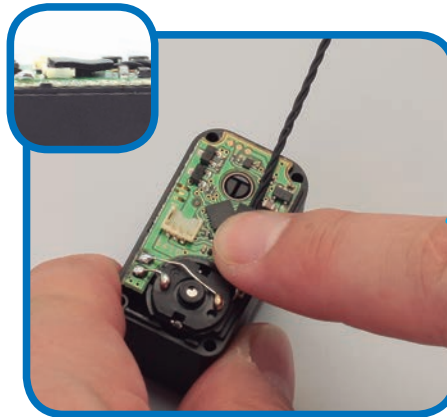
Neck stand (Stage 8)

THE SERVO AND CABLE

TIP!



If you chose not to fit the servo cable's connectors with pads in the previous stage, do so now. Also remember to handle the open servo with great care, as the electronics exposed are very delicate.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side until Step 4. Then gently lift the cover away, taking care not to force it or to damage the circuit board underneath.

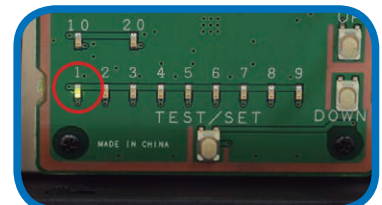
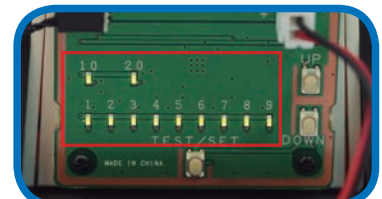
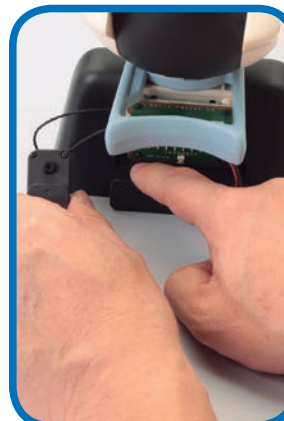
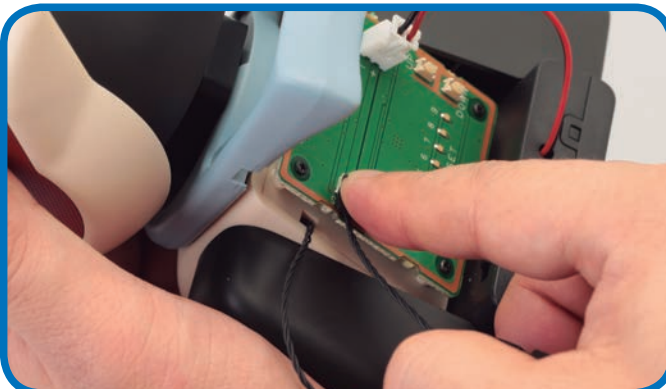
2 Hold the open servo with the circuit board facing upwards. Fetch the servo cable from the previous stage, and press one of its connectors into one of the servo's sockets. Make sure that the connector is sitting flush with the surface of the board (inset).



3 With the padded cable connector securely in place, replace the servo cover removed in Step 1, feeding the cable through the circular hole as you do so.

4 Reattach the cover using the four screws removed in Step 1.

TESTING THE SERVO

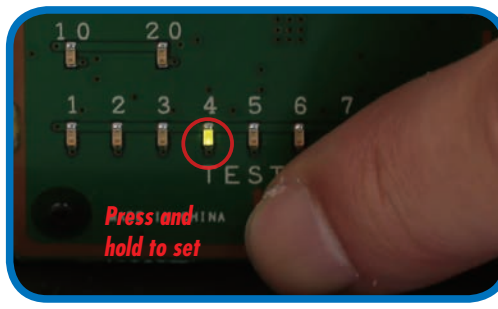
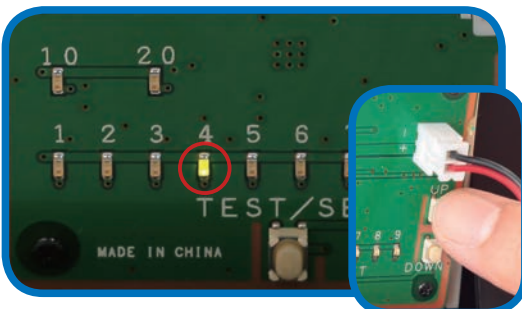


5 Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 2.

6 With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.



- 7** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.



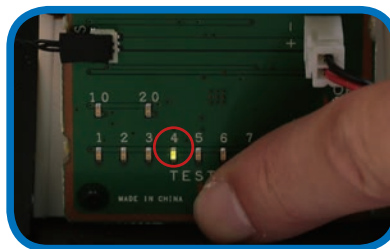
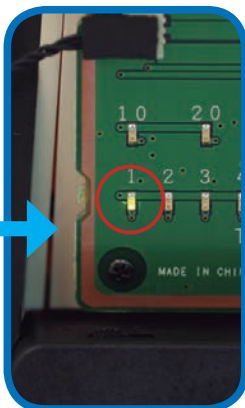
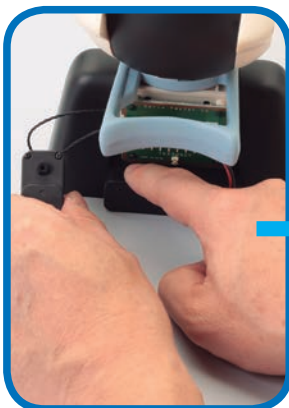
- 8** To set the ID number of the right knee servo to 4, press the UP button three times. Make sure that only the 4 LED is lit, using the UP and DOWN buttons to correct the number if necessary.

- 9** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 4 light should flash a few times, then remain lit. The ID number is now set.

CHECKING THE ID NUMBER



ID=4



- 10** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, switch the power OFF, then back ON. After all the LEDs blink, only the 1 LED should light up.

- 11** To check the ID number, press TEST/SET. Now only the 6 LED should light up. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

Assembled right knee servo

The servo that will operate Robi's right knee is now wired and set with its ID number, ID=4. Store it away safely until next time – it's a good idea to keep it in a sealed plastic bag with the stage number on it for reference.



STAGE 30: ASSEMBLE ROBI'S RIGHT KNEE

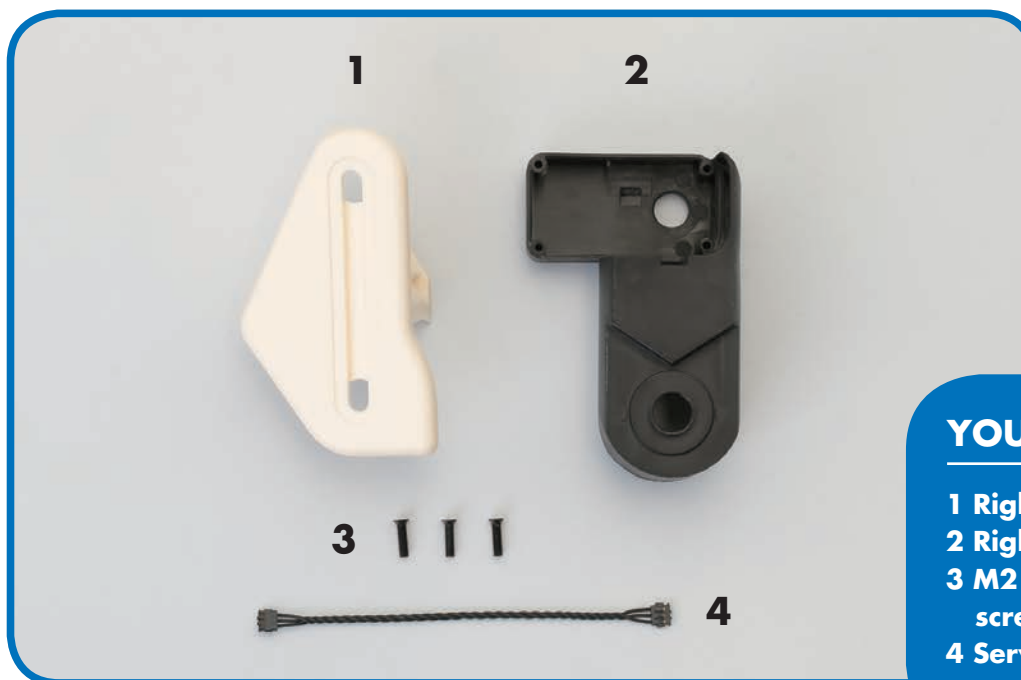
In this session, you will fit the servo prepared in the previous stage – Robi's right knee servo, ID=4 – into the knee frame supplied with this stage. You will then join the knee and ankle frames, before closing the leg with another lower leg panel.

This stage sees you again linking pre-wired servos and handling open servo motors. As ever, great care must be taken when working with the delicate electronic components and

cables, as they are easily damaged. You will also be joining the knee and ankle frames using a D-shaped hole and shaft, so again, take care not to force the part in any way.



PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Right inner lower leg panel
- 2 Right knee frame
- 3 M2 x 7mm countersunk screws x 3 (one is a spare)
- 4 Servo cable 70mm

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



Right foot and lower leg assembly (Stage 28)

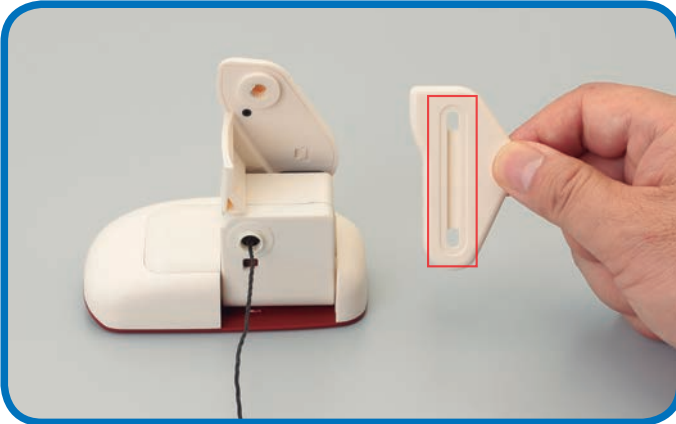


Servo ID=4 (Stage 29)

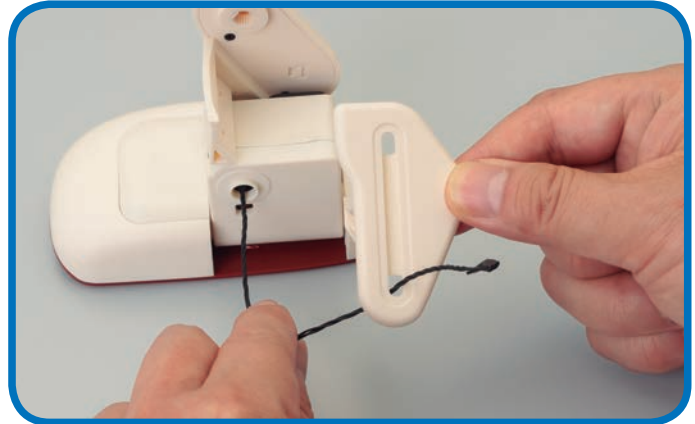


Protective pads (Stage 3)

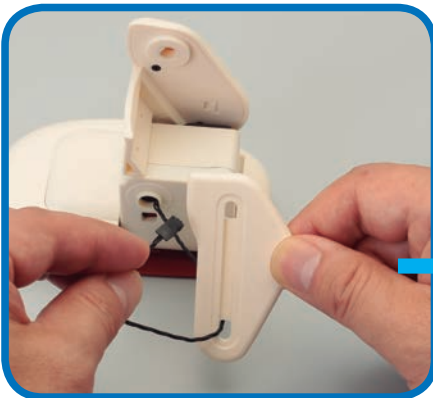
FITTING THE INNER LEG PANEL



- 1 Set Robi's right foot and lower leg assembly out on your work surface, with the ankle perpendicular to the foot's sole. Line up the inner lower leg panel with the outlined groove facing you, as shown.



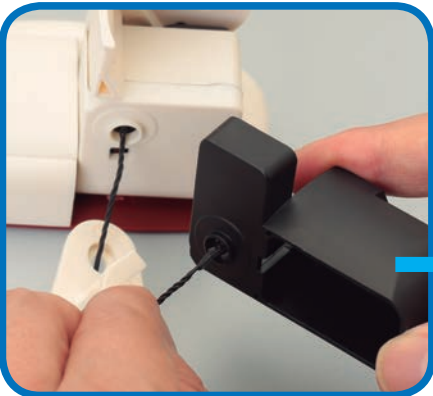
- 2 Feed the servo cable running out from the hole in the ankle frame into the lower hole in the leg panel, from the inside of the part.



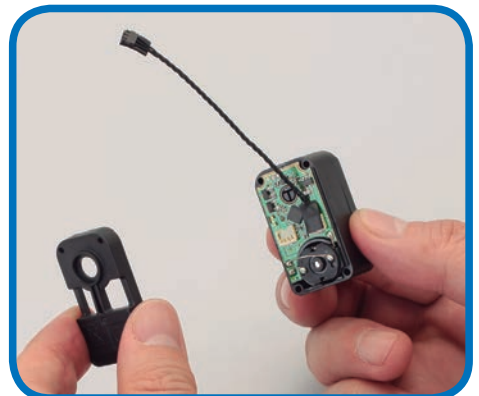
- 3 Now bend the servo cable over and pass its free connector back to the rear of the inner leg panel, by feeding it through the hole at the top of the part.



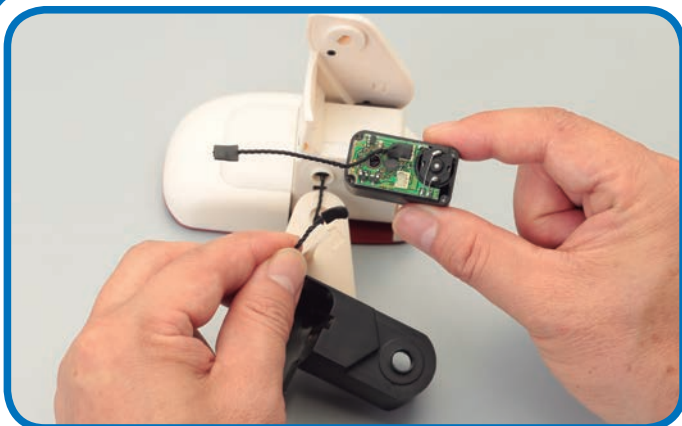
- 4 Now line up the knee frame, orientated as shown above, to the servo cable's connector. The circled hole should face you.



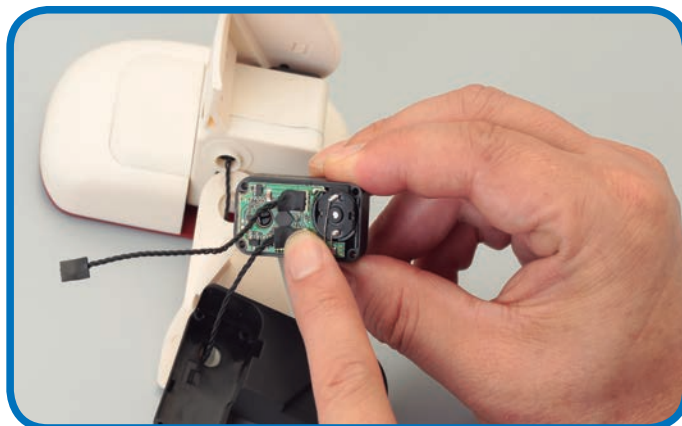
- 5 Feed the cable into the hole and pull it into the inside of the knee frame. Set the assembly down carefully on your work surface, making sure the cable remains in position.



- 6 Take servo ID=4, prepared in Stage 29, and remove its back cover. You will not be using the cover again, but keep the screws for use in Step 12.



- 7** Hold the open servo next to the foot assembly, making sure to hold the servo by its sides and not to disturb any of the exposed electronics or the cable connector.



- 8** Fit the connector of the ankle servo's cable into the free socket on the servo. Again, be careful not to interfere with the servo's components, or dislodge the first connector.



- 9** Making sure both connectors remain securely in their sockets, feed the free end of the 70mm servo cable running from servo ID=4 into the rectangular hole in the knee frame.



- 10** Bringing the servo to the frame, gently pull the cable through the hole and into the knee frame. Try to minimise the slack left between the servo's board and the knee frame by gently pulling on the cable as you bring the parts together.

FITTING THE SERVO



- 11** Press the servo onto the knee frame so that no gaps appear around its edges.



- 12** Use the four screws removed in Step 6 to secure the servo to the frame.

FITTING THE KNEE FRAME



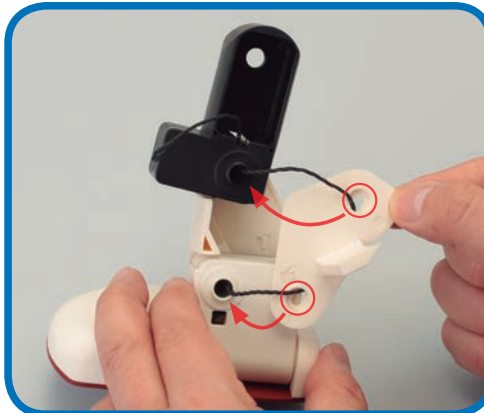
13 Line up servo ID=4, now set into its frame, to the inside face of the ankle frame. The servo's shaft will fit into the D-shaped hole in the frame (circled), with the body of the servo sticking out over Robi's 'shin'. The join is made via another D-shaped hole, so it is important that you align the parts correctly from now on.

Fit the servo so that its shaft is sitting neatly into the D-shaped hole. When the orientation is correct, it will be easy to fit the parts: do not force them!

14



ADDING THE INNER LOWER LEG PANEL



15 Turn the inner lower leg panel to face you – there are two circular indents around each hole. These will accommodate the circular lips on the knee and ankle frames (arrowed).

Fit the leg panel onto the knee and ankle frames.

16



17

Gently pull the middle of the servo cable, as shown above, so that there is no slack inside the leg assembly.



18

Check that no part of the servo cable has been pinched when you fitted the leg panel to the ankle and knee frames. The join should be flush.

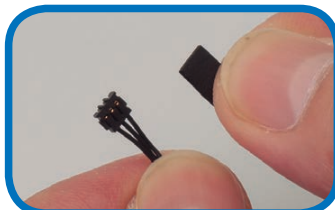
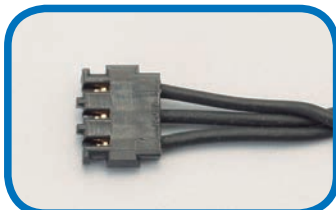


- 19** Locate the two screw holes, circled in the photo above, on the inside of the lower leg panel. Bending the knee forward will help you access these.

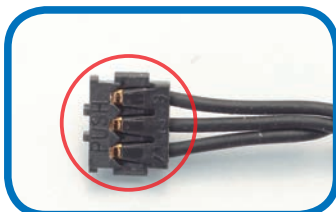


- 20** Use the two M2 x 7mm countersunk screws to secure the parts.

SEALING THE SERVO CABLE



It is advisable to fit the servo cable's protective pads at this stage, but if you want to wait until the cable is used in the build, you can do so. In that case, simply store the cable away safely until it is needed, and remember to apply the pads then.



- 19** As you have done previously, apply a protective pad to the side of this stage's servo cable with PUSH marked on it (circled). Repeat for the other end of the cable.

NOTE: On completion, the length of servo cable sticking out between Robi's ankle and knee will be pulled taut and sit neatly within the groove marked below by the dashed line. Until then, be very careful not to snag or hook this cable on anything, as this will damage the part and result in a great deal of dismantling and rebuilding!



Assembled right foot and lower leg

This stage is now complete, and Robi's right leg is really beginning to take shape. Store all parts away safely until next time.



COMING IN PACK 9

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 31

Prepare Robi's right thigh servo

PARTS PROVIDED

The servo that will act as Robi's right thigh.



Stage 32

Add the hip frame that will join Robi's thigh and hip servos

PARTS PROVIDED

The right inner hip frame and a servo cable.



Stage 33

Prepare Robi's right hip servo

PARTS PROVIDED

The servo that will operate Robi's right hip.



Stage 34

Prepare Robi's left and right 'shoes'

PARTS PROVIDED

The foam and felt parts that make up Robi's padded 'shoes'.